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The C_5 chemistry preceding the formation of polycyclic aromatic hydrocarbons in a premixed 1-pentene flame

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ABSTRACT

The formation of small polycyclic aromatic hydrocarbons (PAHs) and their precursors can be strongly affected by reactions of C_5 species. For improving existing combustion mechanisms for small PAH formation, it is therefore valuable to understand the fuel-specific chemistry of C_5 fuels. To this end, we provide quantitative isomer-resolved species profiles measured in a laminar premixed ($\phi = 1.8$) low-pressure (4 kPa) flame of 1-pentene with photoionization molecular-beam mass spectrometry (PI-MBMS) using tunable synchrotron vacuum-ultraviolet (VUV) radiation. These experimental results are accompanied with numerical simulations, starting from models from the literature by Wang et al. [JetSurF version 2.0 (2010)] and Healy et al. [Energy Fuels 24 (2010) 1521–1528] that were developed for different fuels, but which include 1-pentene as an intermediate, and by Narayanaswamy et al. [Combust. Flame 157 (2010) 1879–1898] focusing on the small PAH chemistry. Taking observed discrepancies between experimental results and simulations into consideration, a mechanism for C_5 chemistry was newly developed including PAH formation pathways, and its performance analyzed in detail. Special emphasis was placed on the initial fuel consumption of 1-pentene as well as on formation pathways of small aromatics. The mechanisms show differences regarding fuel decomposition and hydrocarbon growth reactions. These contribute to noticeable differences between the simulations with different models on one hand, and deviations between model predictions and experimental results on the other. While the new model presents overall satisfactory capabilities to predict the mole fraction profiles of common combustion intermediates, the predictive capability of the literature models was not fully satisfying for some intermediate species, including C_4H_6, C_7H_8, and $C_{10}H_8$. The results indicate that the fuel-specific C_5 reaction routes as well as the mechanism for small PAH formation need further investigation.

1. Introduction

A large number of studies has provided evidence that reactions of C_5 hydrocarbon intermediate species, such as the cyclopentadienyl radical, are expected to play an important role in aromatic ring formation [1–10]. There is, however, a lack of knowledge about the fuel-structure-dependent C_5 chemistry that includes reactions, which contribute to the decomposition of C_5 hydrocarbons, to the formation of the fuel-specific intermediate species pool, and to growth reactions that involve C_5 hydrocarbons as reactant molecules. Involvement of cyclopentadienyl reactions in the formation of aromatic species, including naphthalene in particular, has been noted in combustion and pyrolysis of a number of fuels, including for example cyclopentene [4,5,11], *n*-butane [7], 1,3-butadiene [12] and others. Their importance has been underlined by review articles of McEnally et al. [13] and Wang [14]. In particular, Marinov et al. [2] have shown the relevance of naphthalene production through the reaction step of cyclopentadienyl self-combination using detailed chemical kinetic modeling of premixed flames. It was supported by the seminal work of Melius et al. [3], who have determined the elementary reaction steps in the conversion of two cyclopentadienyl radicals to naphthalene employing quantum chemical calculations. This naphthalene formation pathway was further examined by several groups using experiments, numerical simulations, as well as theoretical calculations. Mebel and Kislov [15], among others, have analyzed the pathways from cylopentadienyl to naphthalene in further detail. More recently,

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Cavallotti and Polino [1] have theoretically identified an additional reaction pathway that starts from two cyclopentadienyl radicals and leads to the formation of a set of azulyl radicals that are, in turn, converted to naphthalene. Understanding the formation pathways of cyclopentadienyl radicals from linear C_5 hydrocarbons is therefore of great interest. 1-Pentene (CH_2=CH-CH_2-CH_2-CH_3) is an interesting representative of the suite of C_5 hydrocarbons because of the double bond in its molecular structure and the possibility to form resonantly stabilized allylic (fuel) radicals during its combustion [16–18]. In general, resonantly stabilized radicals, including C_5H_5, i-C_4H_5, C_3H_5, and C_3H_3, can accumulate in high concentrations and precede the formation of small aromatics and PAHs [13,19–23]. Also, resonantly stabilized radicals are thought to be important drivers in the PAH reactions towards incipient soot [24]. Furthermore, recent results indicate that 1,3-pentadiene, yielded from H-abstraction from the allylic fuel radical of 1-pentene, is an important precursor in the subsequent oxidation to cyclopentadienyl that can give access to naphthalene [12].

Combustion pathways for several C_5 hydrocarbons of decisively different binding structure, including n-pentane and 2-methyl-2-butene, have recently been shown to exhibit significantly different trends of aromatics formation in both non-premixed [17] and premixed laminar flames [16,25], an observation that has led to a closer inspection of the respective fuel-specific precursor pool and their reactions [16,17,25]. Regarding 1-pentene, several earlier studies have examined its combustion kinetics experimentally and with numerical models for different combustion conditions, including premixed flames, shock tubes and rapid compression machines [17,26–35]. Most of these investigations have not focused on the formation of small PAHs and their precursors. Some formation pathways of aromatic species were discussed in Ref. [27], without providing isomer-resolved species profiles, however.

The present work aims to discuss the C_5 chemistry preceding PAH formation in a laminar premixed low-pressure 1-pentene flame ($\phi = 1.8$, 4 kPa). First, we provide new isomer-resolved data measured with photoionization molecular–beam mass spectrometry (PI-MBMS) using tunable synchrotron vacuum ultraviolet (VUV) radiation. Second, we have developed a new mechanism for the combustion of C_5 hydrocarbons that describes the chemistry up to the formation of small PAH molecules. This model was then used to simulate the present 1-pentene flame and to compare its performance against the experimental data as well as to different well-validated models for C_5 and PAH chemistry from the literature [36–38]. The simulations revealed substantial differences between the models and showed an overall satisfactory performance of the new model. Our study is expected to provide valuable information for further mechanism development of hydrocarbon combustion mechanisms that include C_5 sub-mechanisms.

2. Experimental procedure

A fuel-rich laminar premixed 1-pentene flame was investigated with PI-MBMS in the low-pressure flame apparatus at the Chemical Dynamics Beamline at the Advanced Light Source of the Lawrence Berkeley National Laboratory [39,40]. The flame was stabilized on a 6 cm diameter McKenna burner at 4 kPa and had a cold-gas composition of 1-pentene/oxygen/argon (14.5%/60.5%/25%) with a mass flow rate of 0.00312 g cm^{-1} s^{-1}. Gas flows were controlled by calibrated mass flow controllers. Purities for oxygen and argon were 99.999% and ≥99% for 1-pentene. The flame conditions were chosen to be similar to preceding studies of different C_5 fuels to enable an approximate comparison of the general reaction behavior [16,25].

The setup consists of a low-pressure flame chamber, a two-stage differentially pumped vacuum chamber, and a time-of-flight mass spectrometer. Flame gases were extracted *via* a quartz cone and expanded into the first pumping stage of the vacuum chamber (10^{-2} Pa) to form a molecular beam that preserves the sample composition and enables the detection of reactive species. The molecular beam was guided through a skimmer into the ionization chamber (10^{-4} Pa) where the molecules were crossed and ionized with the tunable VUV radiation ($\Delta E = 0.05$ eV). The generated ions were extracted into a time-of-flight mass analyzer, energetically focused by a reflectron, and then detected using a micro-channel plate and counted with a multi-channel scaler. The instrument enables the detection of species with concentrations as low as ~1 ppm, and the mass-resolution of $m/\Delta m \approx 4000$ allows for the separation of the exact C/H/O elemental composition of each species. Species profiles as function of the height above the burner were recorded using photon energies between 8.7 eV and 16.65 eV to minimize undesired fragmentations. The photoionization efficiency (PIE) curves, which were recorded in the reaction zone of the flame, allow for the separation of isomeric intermediates through identification of their ionization thresholds. Because MBMS is an invasive technique, in which the flow and temperature fields of the laminar flame are perturbed by a sampling probe [41,42], a perturbed temperature profile was determined and used as input parameter for kinetic modeling. The perturbed temperature profile of the flame was obtained based on the relationship between the flame temperature and the pressure in the first pumping stage following a procedure described in Ref. [42], using a calibration point from the exhaust gas temperature. This calibration point was measured using laser-induced fluorescence of OH radicals as described in Ref. [43] and an exhaust gas temperature of 2000 (±150) K was experimentally obtained.

Data evaluation procedures have been reported previously [44,45] and are therefore not repeated here. The experiment itself as well as the data evaluation procedure cause experimental uncertainties. The uncertainties of the obtained mole fractions depend on several factors, and a detailed discussion of the typically associated errors is given in Ref. [40]. For the major species, uncertainties are ~20–30%; for intermediate species, uncertainties depend on the quality of the photoionization cross sections [46], and the error can be as large as factors of 2–4.

3. Kinetic modeling

Numerical simulations were performed using the 1D premixed flame module in the FlameMaster code [47]. The conservation equation of mass and the species balance equations were used to formulate the numerical problem with the disturbed experimental temperature profile as input parameter replacing the energy equation. A detailed description of the module can be found in Ref. [48]. A minimum number of 300 grid points was employed in the simulation for a good accuracy of the calculations. It was checked that further addition of grid points barely affects the accuracy of the results. The cold gas velocity was determined as 0.4223 m/s based on the condition of 4 kPa and 298 K. The numerical simulations in this study were performed with three different models that all contain sub-mechanisms for base, C_5, and PAH chemistry. Figure 1 gives a general overview of how the different sub-mechanisms from the literature were combined in each model, and a detailed description is provided in the following subsections.

3.1. Numerical simulations with literature models

In a previous study by Cheng et al. [29] regarding ignition delay times at high temperatures, four 1-pentene mechanisms [29,33,36,37] were compared in terms of their performance. Here, two of them, referred to as "USC" [36] and "NUIG" [37] were considered because of their direct availability (see Fig. 1). As shown in Ref. [29], both mechanisms predict the ignition delay times of

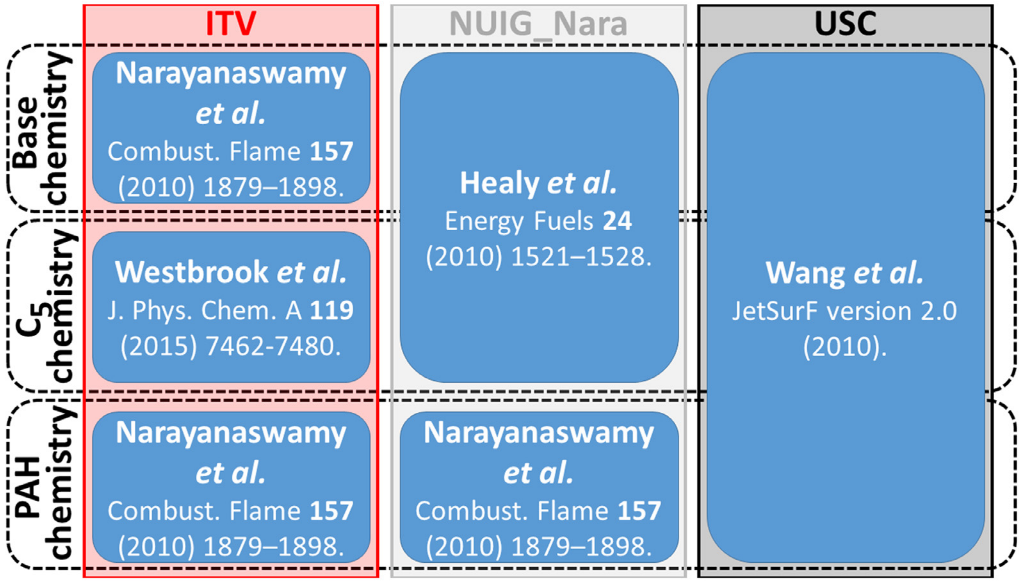

Fig. 1. Illustration of the chemical kinetic models (ITV, NUIG_Nara, and USC) used in this study. Different sub-mechanims for base, C_5 as well as PAH chemistry from literature were combined in order to describe the high-temperature oxidation of 1-pentene up to the formation of small PAH molecules.

1-pentene with reasonable accuracy. However, they differ strongly in terms of their reaction pathways, as illustrated in Fig. 4 of this study. A comparison between these two models can thus provide additional insights into the dominant reaction pathways of a 1-pentene flame.

The USC mechanism includes the 1-pentene thermal decomposition, H-atom abstraction by H-radical forming three 1-pentenyl radicals, detailed decomposition channels of 1-pentenyl radicals [49], and chemically activated hydrogen addition reactions of 1-pentene. Hydrogen abstraction reactions from 1-pentene by O, OH, O_2, HO_2, and CH_3 are included for the formation of 1-penten-5-yl radicals, but not for 1-penten-3-yl and 1-penten-4-yl radicals. Furthermore, the USC mechanism comprises reaction channels involving several aromatic species, such as benzene, phenol, and toluene.

The NUIG mechanism contains the reactions of 1-pentene as part of the *n*-pentane oxidation chemistry, and these elementary reactions are derived based on the concept of reaction classes and rate rules [37]. Formation pathways of PAHs are, however, missing in the NUIG model. To facilitate the simulation of 1-pentene flames with inclusion of small PAHs and their precursors, the NUIG model was extended by the well-validated PAH chemistry of Narayanaswamy et al. [38] taken from the literature (see Fig. 1). The model of Narayanaswamy et al. [38] was developed to jointly describe the oxidation of substituted aromatic species and the formation and growth of PAHs and includes detailed PAH formation pathways up to species with four aromatic rings. As shown in Fig. 1, the PAH sub-mechanism of this model was adapted and incorporated into the NUIG model and the resulting model is referred to as “NUIG_Nara”.

3.2. Mechanism development

As it will be shown in Section 4, the predictive capabilities of the USC and NUIG_Nara models are not fully satisfactory, which motivates the development of a new high-temperature 1-pentene mechanism, in the following referred to as “ITV”. The mechanism is developed based on the mechanism of Narayanaswamy et al. [38], which also includes a well-validated C_0-C_4 base mechanism [50]. The fuel-specific reactions of 1-pentene and their reaction rate parameters are taken from Refs. [18,36,37]. As 1-pentene is the direct product of the C−H β–scission of pentyl radicals, the relevant reactions of *n*-pentane, *i.e.* the decomposition and H-abstraction reactions of *n*-pentane, were incorporated into the present mechanism for completeness. The reactions were taken from Westbrook et al. [18] and pressure-dependent rate parameters were used, if available. The resulting model was validated using the new species data, which will be shown below. For further validation, the proposed model was compared to experimental data of burning velocities available in the literature and the results are shown in Fig. S1 in the Supplemental Material SM1. The most significant modifications to the mechanism are presented as follows:

(1) Recent studies on the oxidation of 1-butene [51] highlighted the importance of the chemically activated H-atom addition reaction to 1-butene forming directly two small molecules in the premixed laminar flame. Thus, the chemically activated H-atom addition reactions were incorporated in the present mechanism for 1-pentene as well, and their rate constants were specified according to the analogous reactions of 1-butene. These reactions were found in this study (see Section 4.2) to be very important consumption channels of 1-pentene.
(2) 1,3- and 1,4-pentadienes are formed in the 1-pentene flame from the decomposition of pentenyl radicals. In the USC mechanism [36], they proceed *via* 1,4-pentadien-3-yl radicals to form cyclopentadiene directly. Through further H-abstraction, cyclopentadienyl radicals are produced and can react with methyl radicals yielding fulvene and subsequently benzene. To improve the model prediction for PAH formation, the reaction paths from 1,3- and 1,4-pentadienes to cyclopentadiene, the interaction of 1,4-pentadien-3-yl, 1,3-pentadien-5-yl, and cyclopentenyl radicals, and their reactions to cyclopentadiene were incorporated. The corresponding reactions were adopted from the Aramco 3.0 mechanism [52].
(3) The mechanism of Narayanaswamy et al. [38] was developed originally for the combustion of engine relevant fuels and its C_0-C_4 and PAH chemistry was validated extensively at engine-relevant high-pressure conditions. As the present measurements were performed at a much lower pressure of 4 kPa, pressure-dependencies of rate constants were added for reactions that are sensitive towards the prediction of flame data. For example, Miller and Klippenstein [53] calculated the rate

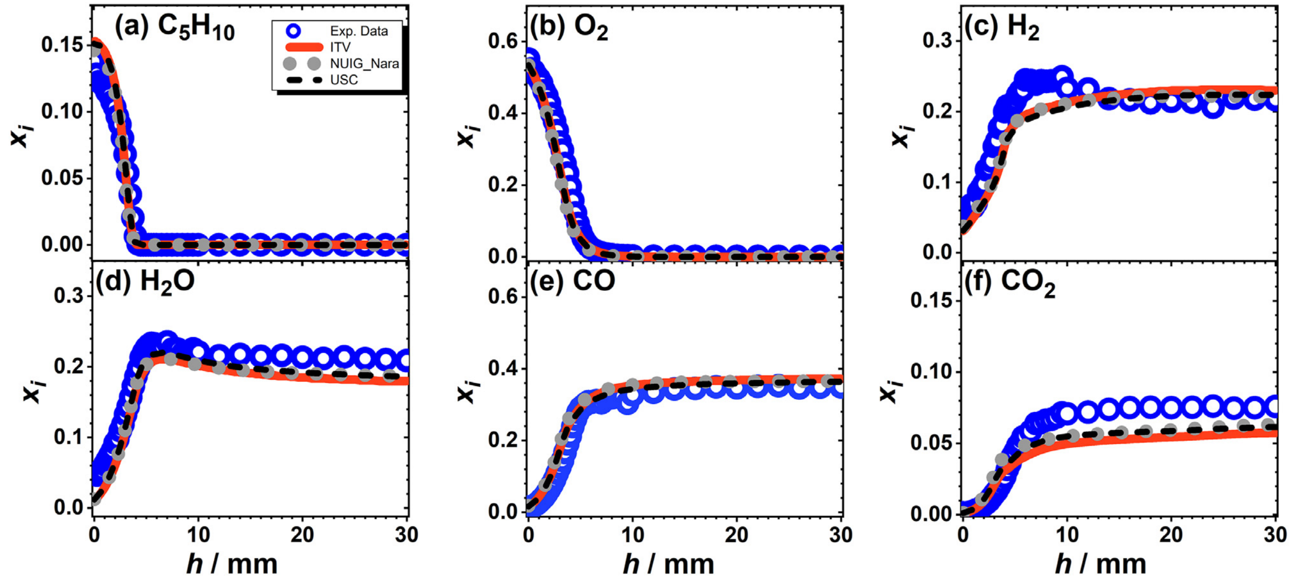

Fig. 2. Major species (C_5H_{10}, O_2, H_2, H_2O, CO, and CO_2) mole fractions x_i as a function of the height above the burner h; symbols: experimental data; solid, dotted and dashed lines: simulations with the ITV, NUIG_Nara, and USC models, respectively. Note that the experimentally detected H_2 signal, from which the respective mole fraction profile was determined, was smoothed (details are provided in Fig. S3 of the Supplemental Material SM1).

constants for the recombination reactions of hydrogen and propargyl radical at various pressures. While the mechanism of Narayanaswamy et al. [38] only considers the rate constants at 10 bar, the pressure-dependent rate constants are incorporated in the new ITV mechanism. For the recombination reactions of H with propyne or allene to propenyl isomers, the pressure-dependent rate constants from Miller et al. [54] were employed. These reactions are mainly related to the C_3 chemistry involving reactions of propenyl isomers, propyne, allene, and propargyl radical, which are important for the formation of benzene.

The reaction mechanism, the thermodynamic properties, and the transport data of the ITV model are available in the Supplemental Material SM2 (SM2_Mech, SM2_Thermo, SM2_Trans).

4. Results and discussion

Experimental and simulated mole fraction profiles of the main species and selected intermediate species are provided in the following discussion and PIE curves for selected species are additionally shown to identify the respective main isomer. Maximum mole fractions for all quantified species and the corresponding modeling results are given in Table S1 in the Supplemental Material SM1. The post-processed data, *i.e.* the corresponding data to the mole fraction profiles of all quantified major and intermediate species, are available in the Supplemental Materials SM3 and SM4, respectively, while the data of the temperature profile of the 1-pentene flame is provided in the Supplemental Material SM5.

4.1. Validation of the mechanisms

The mole fraction profiles of the major species (C_5H_{10}, O_2, H_2, H_2O, CO, and CO_2) are presented in Fig. 2 together with the predictions of the ITV (solid line), NUIG_Nara (dotted line) and USC mechanisms (dashed line). The mole fraction profiles of Ar as well as the perturbed temperature profile are presented in Fig. S2a-b in the Supplemental Material SM1. Experimental and modeling results with all three mechanisms show reasonable agreement for the major species. Minor discrepancies between the simulated and experimental mole fractions are observed near the burner exit, especially for H_2. Note that the detection of H_2 close to the burner is challenging with the PI-MBMS set-up. Because of the low signal-to-noise ratio, the experimentally detected H_2 signal has been smoothed and the original H_2 signal as well as a detailed description of the smoothing procedure are provided in the Supplemental Material SM1 (Fig. S3). Experimental errors for H_2 can therefore be significantly larger than for the other major species.

Experimental and simulated mole fraction profiles of CH_4, C_2H_2, and CH_2O are shown in Fig. 3 to verify the predictive capabilities of the mechanisms. For methane (CH_4) and acetylene (C_2H_2), reasonable agreement between experiment and modeling regarding the profile shape and maximum mole fraction is observed for all models (Fig. 3a–b). Only for formaldehyde (CH_2O), larger deviations between the experiment and the simulations with the ITV and USC models are observed, which are, however, still within the experimental uncertainty. Note that mole fraction profiles of further species will be discussed below. Besides the mole fraction profiles for the 1-pentene premixed flame that are presented in this study, the newly developed ITV model was also validated using experimental data of burning velocities available in the literature and the results are shown in Fig. S1 in the Supplemental Material SM1. The three models are sufficient for analyzing the combustion process of 1-pentene, because all of them provide a respectable description of the global combustion chemistry of 1-pentene including speciation data as well as flame speeds.

4.2. Initial fuel consumption pathways

The first step in the fuel consumption depends on the molecular fuel structure and it can be initiated by different reaction types such as H-atom abstraction, fuel dissociation, and isomerization reactions as well as radical addition reactions in the presence of unsaturated fuels. Information about the initially formed fuel consumption intermediates are crucial to understand the fuel-structure-dependent reaction kinetics because these intermediates are at the origin of the subsequent reactions including molecular growth. Figure 4 shows a fuel oxidation scheme of 1-pentene with net percentage consumption of each species for the ITV (top), NUIG_Nara (middle), and USC (bottom) models, obtained by the respective flux analyses that were integrated from h = 0–30 mm. Only selected reaction pathways with contributions of ≥2% for

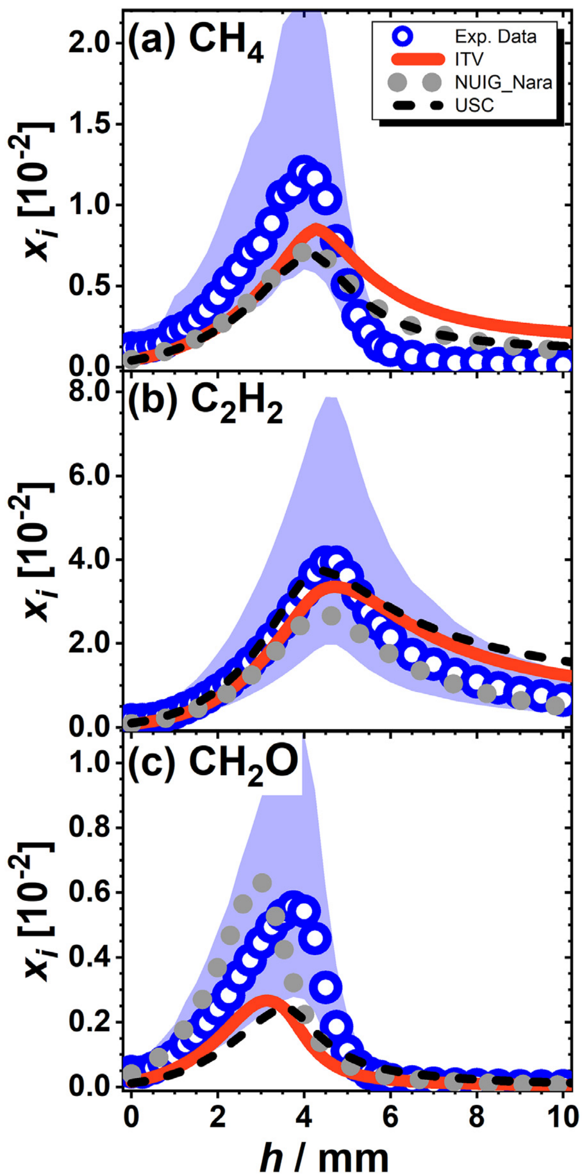

Fig. 3. Mole fractions x_i as function of the height above the burner h of (a) CH_4, (b) C_2H_2, and (c) CH_2O. Symbols: experimental results; solid, dotted and dashed lines: simulations with the ITV, NUIG_Nara and USC models, respectively. Experimental uncertainties (factor of 2) are visualized as shaded area.

initial and ≥5% for secondary decomposition steps are included. The flux analyses with the ITV, NUIG_Nara, and USC models were performed for the experimental conditions that were presented in Section 2.

From the flux analyses, it becomes obvious that the primary consumption reactions vary remarkably between the different models. The H-atom initiated fission of the allylic C–C bond that leads to the formation of propene (C_3H_6) and the ethyl (C_2H_5) radical, is the dominant decomposition reaction in the newly developed ITV model (37%). While this reaction has, with 17%, also a strong contribution in the USC model, it is not present in the NUIG_Nara model. Both, the NUIG_Nara and the USC model were not developed and validated specifically for 1-pentene. Nevertheless, the high-temperature oxidation of C_5 hydrocarbons, *i.e. n*-pentane, is included in both mechanisms. Hence, it was assumed that both well-validated literature mechanisms should be capable to predict the high-temperature reaction behavior of 1-pentene, because this molecule is a linear alkene and its reactions are already incorporated in the sub-mechanism of *n*-pentane. Both mechanisms were therefore used in this work to discuss the relevance of different reactions. In the USC model, the direct formation of C_3H_5 and C_2H_5 radicals *via* thermal dissociation of the allylic C–C bond is, with 36%, the most important consumption pathway of 1-pentene. It is also the third-most significant decomposition reaction in the ITV and NUIG_Nara mechanisms with contributions of 13% and 11%, respectively. These pathways can be considered as chemically favored by the binding situation of 1-pentene with its allylic and alkylic C–C and C–H bonds. The importance of breaking the allylic bond was also shown by Hansen et al. [55] in the modeling of a 1-hexene flame.

For the NUIG_Nara model, the most important consumption pathway is, with 36%, the H-atom addition to the double bond of 1-pentene forming the 2-pentyl radical (C_5H_{11}-2). In this model, the formation of the 1-pentyl radical (C_5H_{11}-1) *via* H-atom addition to the double bond contributes also with 10% to 1-pentene's decomposition. This is in strong contrast to the results with the new ITV model, where both the reaction pathway *via* the 2-pentyl radical (C_5H_{11}-2) and *via* the 1-pentyl radical (C_5H_{11}-1) are of minor importance with 6% and 1%, respectively. Both aforementioned reactions are negligible in the USC model with contributions of <1%.

For all three models, the formation of the allylic fuel radical C_5H_9-3 is the second-most significant consumption pathway with 19%, 22%, and 26% for the ITV, NUIG_Nara, and USC models, respectively. The formation of allylic radicals is thermodynamically favored, because the bond dissociation energy of allylic bonds is lower than for the alkylic and vinylic ones. Moreover, allylic radicals are resonantly stabilized because of three vicinal sp^2-hybridized carbon atoms in the allylic group.

The large differences in the initial decomposition steps lead to divergences in the predicted mole fraction profiles of the intermediates. The following discussion therefore focuses on the formation of primary intermediates and their subsequent reactions. Because the initially formed radicals are not detectable in our experiment, experimental information on initial reaction pathways is provided from stable intermediates formed in the early steps of the fuel decomposition. The mole fraction profiles of selected intermediate species (C_2H_4, C_3H_4, C_3H_6, C_4H_6 and C_5H_8) are presented in Figs. 5 and 6. Some of these intermediates, which are highlighted in bold font and boxes in the oxidation scheme (Fig. 4) could already be formed from initial reaction steps. According to the flux analysis performed using the ITV mechanism, however, >70% of C_3H_6, >80% of C_4H_6, and 100% of C_5H_8 are produced *via* the pathways included in Fig. 4.

As illustrated in Fig. 4, ethene is an important intermediate in the combustion process of 1-pentene that can be formed directly or *via* primary intermediates including C_2H_5 ($C_2H_5 \rightleftharpoons C_2H_4$+H), C_3H_6 (C_3H_6 + H $\rightleftharpoons C_2H_4$+$CH_3$), C_4H_6 (C_4H_6+H $\rightleftharpoons C_2H_4$+$C_2H_3$), and C_5H_9-5 (C_5H_9-5 $\rightleftharpoons C_2H_4$+$C_3H_5$). According to the ITV model, however, only 7% of C_2H_4 are formed *via* initial reactions and thus later reaction steps are mainly responsible for ethene formation, even though Fig. 4 creates the impression that C_2H_4 is a direct fuel decomposition product. The model prediction is in accordance with the experiment that shows the peak mole fraction of C_2H_4 at h = 4.00 mm, while the peak mole fraction of other initially formed intermediates such as C_4H_6 and C_5H_8 can be observed at slightly earlier reaction times (h = 3.75 mm), respectively. The experimental maximum mole fraction of C_2H_4 of 5.1×10^{-2} is underpredicted by all three models by approximately a factor of two (Fig. 5a). Ethene mainly reacts further *via* the C_2H_3 radical to acetylene (C_2H_2) that is known to be an important precursor for the formation of small PAHs. As already shown in Fig. 3b, the maximum mole fraction of C_2H_2 of 3.9×10^{-2} is reached at h = 4.75 mm indicating that this species is dominantly formed in late reaction steps.

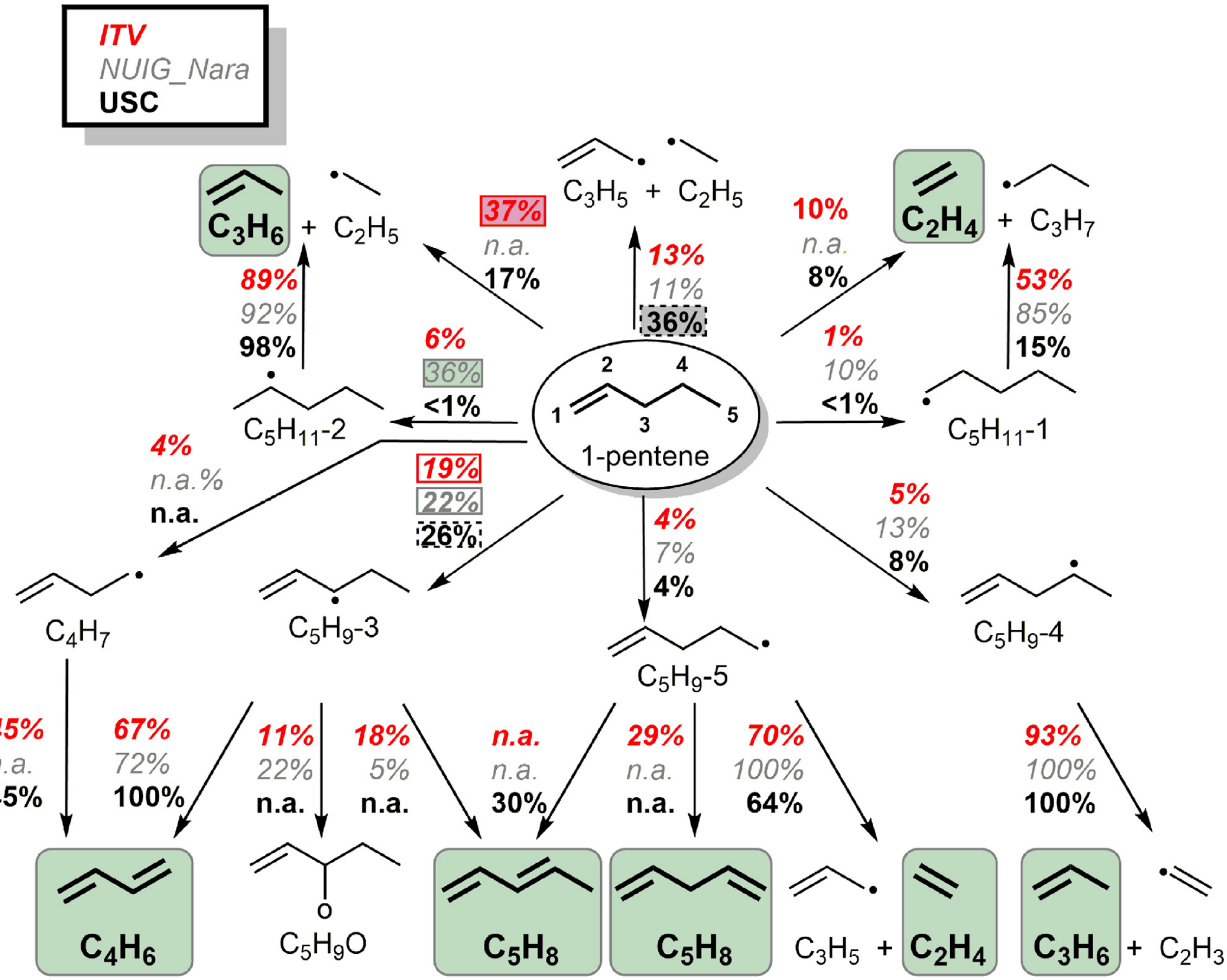

Fig. 4. Fuel oxidation pathways of 1-pentene with contributions of ≥2% for initial and ≥5% for secondary decomposition steps. Net consumptions from the reaction flux analyses (integrated from $h = 0$–30 mm) are shown in percentages next to the arrows for the ITV (top), NUIG_Nara (middle), and USC (bottom) models. The boxes around the percentages highlight the consumption pathways with the first (shaded) and second (open) largest contribution for the respective models. Reactions labeled with "n.a." are not present in the respective model. Mole fraction profiles are provided for species that are highlighted in bold font and boxes. Please note that not all decomposition steps are shown for clarity.

Propene has been found in the flux analyses to be the most important primary product and this prediction is in good agreement with the comparatively high peak mole fraction of 2.0×10^{-2} that was experimentally detected at a height above the burner of $h = 3.75$ mm, *i.e.* formed in initial fuel decomposition reactions (see Fig. 5b). As shown in Fig. 4, C_3H_6 is mainly produced *via* the chemically activated H-atom addition reaction for 1-pentene, when using the ITV mechanism. This reaction was not included in the NUIG mechanism where C_3H_6 is produced by the decomposition of 2-pentyl radical that is the product of the sequential H-addition to 1-pentene. While the NUIG mechanism predicts the maximum concentration of C_3H_6 with reasonable accuracy, the shape of the profile is not well captured. The USC mechanism contains this chemically activated reaction, but predicts a low contribution for it, thus leading to an under-prediction of the concentration of C_3H_6. By incorporating the chemically activated H-atom addition reactions for 1-pentene and using the calculated rate constants of the analogous reactions of 1-butene, the new ITV mechanism predicts the propene concentration very well and shows a better agreement with the experimental data than the literature mechanisms. In the investigated flame, propene further decomposes *via* H-addition and fission of the C–C bond to $C_2H_4 + CH_3$ or *via* H-atom abstraction to the allyl radical (C_3H_5) through which allene (A-C_3H_4) and propyne (P-C_3H_4) are produced. From the subsequent reactions of allene and propyne, propargyl radicals are produced, which recombine to fulvene and benzene in the reaction zone. The experimental and numerical mole fraction profiles of allene and propyne are compared in Figs. 5c-d. Good agreement is observed between the experimental data and the ITV mechanism. The predictions of allene and propyne were found to be sensitive to the pressure-dependence of their reaction rate constants. The incorporation of the pressure-dependent rate constants improved the model performance.

Both, C_4H_6 and C_5H_8 (mainly 1,3-butadiene and 1,3-pentadiene; PIE curves: Fig. 6c-d) can be formed *via* the most abundant fuel radical C_5H_9-3, which results from the third-most significant consumption pathway of 1-pentene as predicted by all three models (Fig. 6a-b). The experimentally observed large concentrations of 1.0×10^{-2} for C_4H_6 and 1.1×10^{-3} for C_5H_8 provide evidence for the relevance of these reaction pathways. The detected mole fraction of C_4H_6 is an order of magnitude higher than that of C_5H_8 indicating that the allylic fuel radical preferably decomposes *via* C–C β-scission to form 1,3-butadiene instead of producing 1,3-pentadiene *via* C–H β-scission. From a chemical point of view, the formation of C_4H_6 is favored because the bond dissociation energy of allylic C–C bonds is with ~262-365 kJ/mol slightly lower than that of

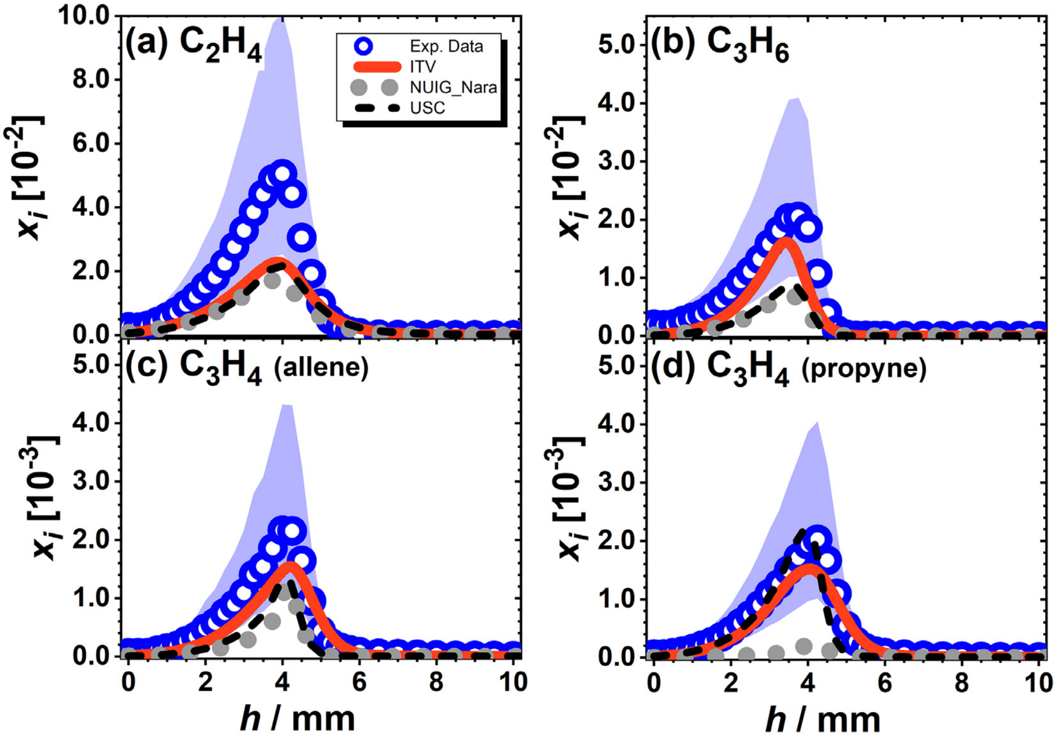

Fig. 5. Mole fractions x_i as function of the height above the burner h of (a) C_2H_4, (b) C_3H_6, (c) C_3H_4 (allene), and (d) C_3H_4 (propyne). Symbols: experimental results; solid, dotted and dashed lines: simulations with the ITV, NUIG_Nara and USC models, respectively. Experimental uncertainties (factor of 2) are visualized as shaded area.

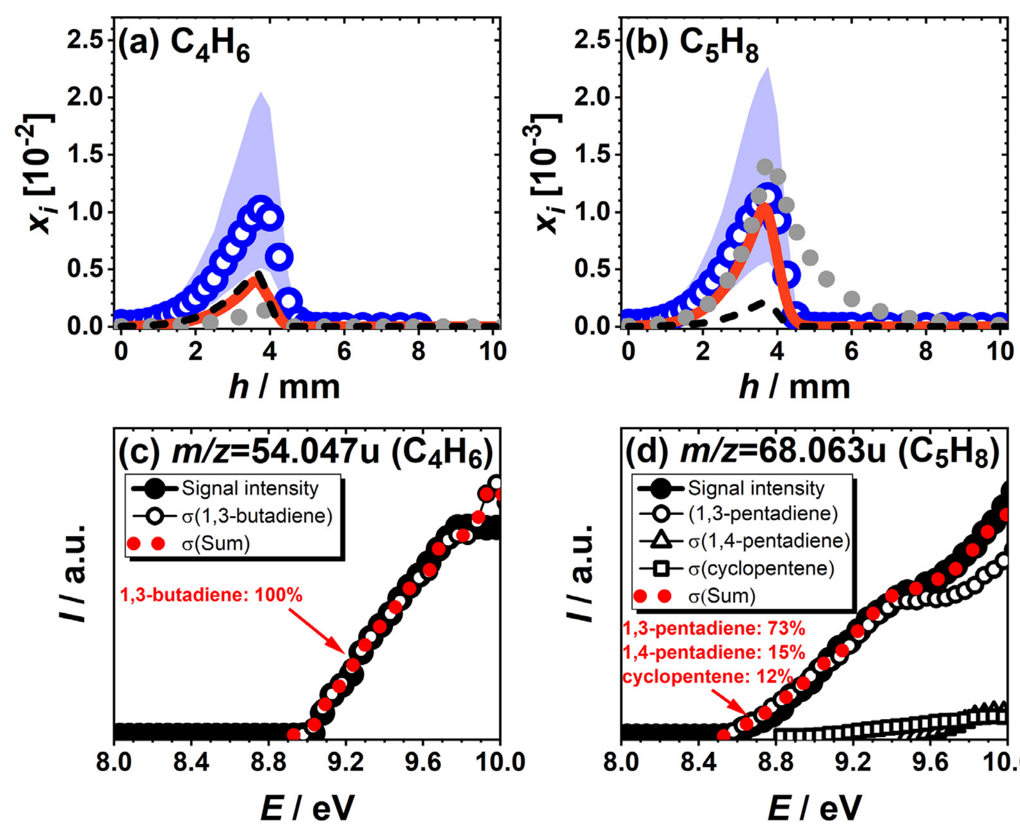

Fig. 6. Mole fractions x_i as function of the height above the burner h of (a) C_4H_6, and (b) C_5H_8. Symbols: experimental results; solid, dotted and dashed lines: simulations with the ITV, NUIG_Nara and USC models, respectively. Experimental uncertainties (factor of 2) are visualized as shaded area. Flame-sampled photoionization efficiency curves as function of the photon energy E for (c) $m/z = 54.047$ u (C_4H_6) and (d) $m/z = 68.063$ u (C_5H_8). Closed symbols (connected to guide the eyes) represent the measured signal, open symbols the tabulated photoionization cross section curve of the respective isomers, and the obtained sum signal is shown as dotted line. The contributions of the main isomers are shown as percentages.

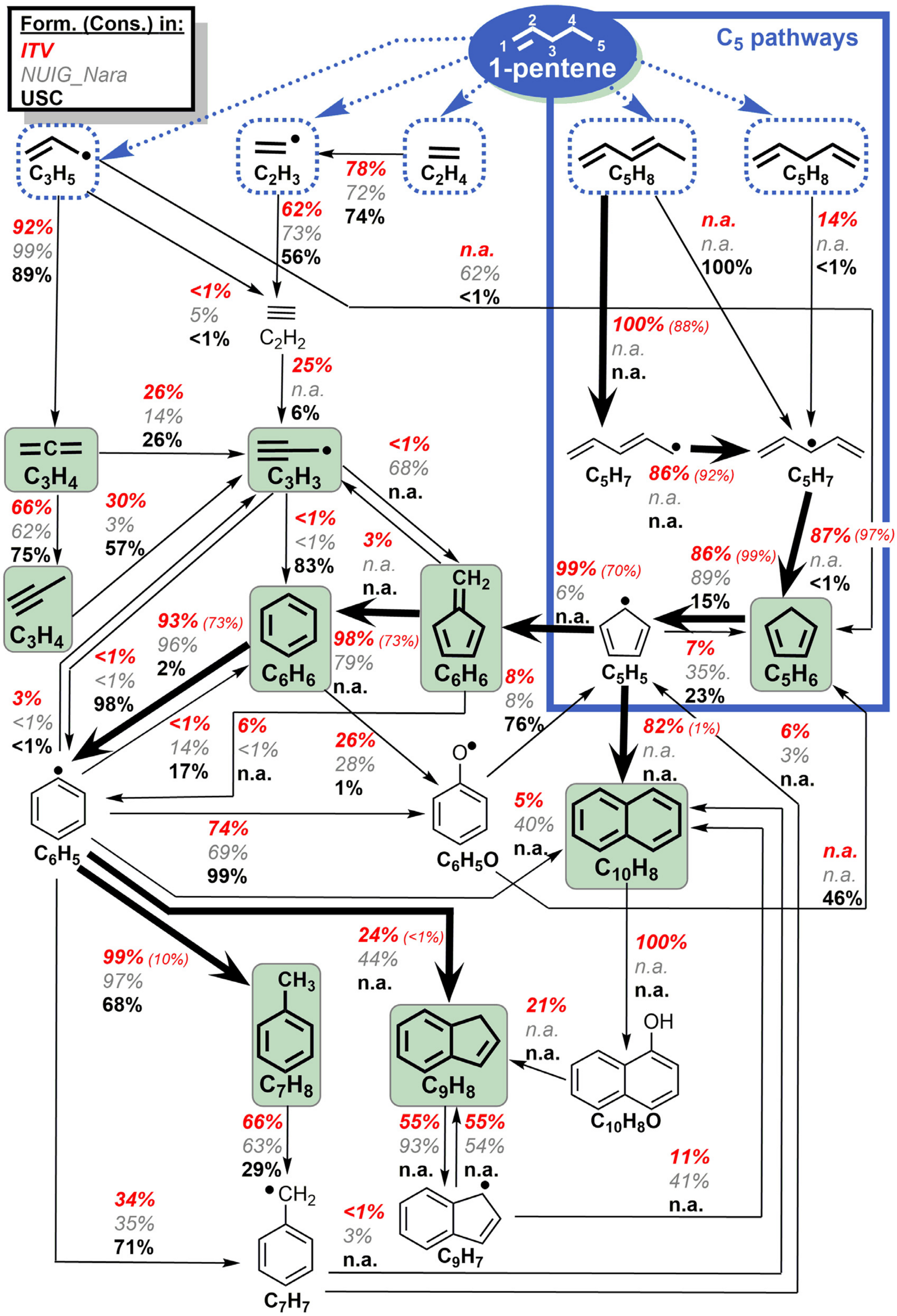

Fig. 7. Formation of aromatic species: reaction sequences starting from initial decomposition products (dotted boxes). Net formations from the reaction flux analyses (integrated from $h = 0$-30 mm) are shown in percentages next to the arrows for the ITV (top), NUIG_Nara (middle), and USC (bottom) models. Bold arrows highlight dominant formation pathways in the ITV model, and the respective net consumptions are given in percentages in parentheses. Reactions labeled with "n.a." are not present in the respective model. Mole fraction profiles are provided for species that are highlighted in bold font and boxes. Note that not all formation steps are shown for clarity.

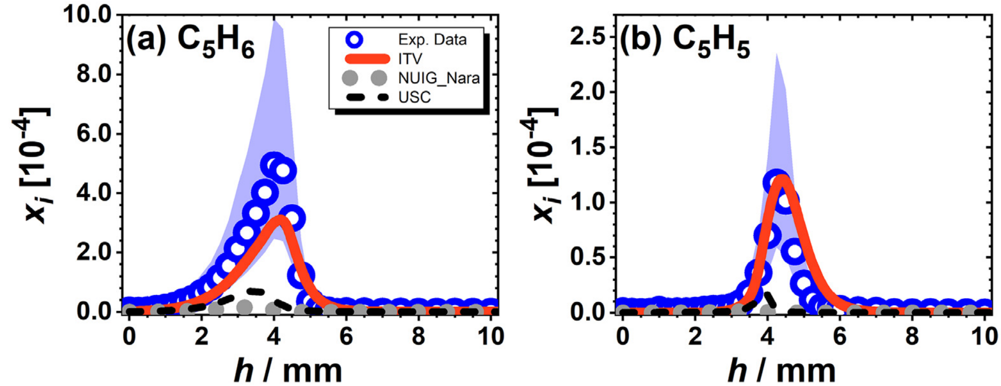

Fig. 8. Mole fractions x_i as function of the height above the burner h of (a) C_5H_6, and (b) C_5H_5. Symbols: experimental results; solid, dotted and dashed lines: simulations with the ITV, NUIG_Nara and USC models, respectively. Experimental uncertainties (factor of 2) are visualized as shaded area.

allylic C−H bonds (~372 kJ/mol), and CH_3 abstraction is therefore preferred over H-atom abstraction [16,17,56]. The experimental observation is consistent with the results from all three models that show major contributions of the reaction pathway leading to C_4H_6. The decomposition *via* 1,3-pentadiene contributes only with 5% in the NUIG_Nara and 18% in the ITV models, whereas this pathway is not considered in the USC mechanism. The difference in the reaction flux leads to different numerical results for C_4H_6 and C_5H_8, as seen in Fig. 6. The present ITV mechanism predicts the profiles of C_5H_8 very accurately, while the NUIG_Nara and USC mechanisms over- and under-predict the concentrations, respectively. For butadiene, the predictions of the ITV and USC mechanisms are within an experimental uncertainty of a factor of 4.

In summary, all models can predict 1-pentene's global combustion chemistry satisfactorily, but remarkable differences exist already in the first initial fuel consumption steps. A plausible prediction of the initial decomposition steps is a prerequisite for a model to reasonably capture the significance of subsequent reaction steps, *i.e.* the fuel-specific combustion chemistry. Differences in the prediction of the initial fuel decomposition steps lead therefore to noticeable differences in the predicted profiles for intermediate species, and by comparing the numerical results from the different models with experimental validation data, it is possible to assess whether the predictive capabilities of one model might be better for the conditions in question compared to another model. The importance of model comparisons for the assessment of their predictive capabilities was confirmed by results of Hansen et al. [57], who have shown that different models, with overall similar predictive capabilities predict different reaction pathways to be important.

4.3. Formation of small PAHs and their precursors

The formation pathways of aromatic species are of interest to gain a better understanding of PAH and soot formation processes. Figure 7 depicts major growth sequences contributing to the formation of aromatic hydrocarbons that are starting from the initial decomposition products (dotted boxes) mentioned earlier and already shown in Fig. 4. These reaction pathways were identified by using reaction flux analyses from the different models, which are based on the formation pathways of the respective species that were integrated from $h = 0$–30 mm. Formation pathways that are, according to the ITV model, of significant relevance are highlighted in bold. Modeling results from the newly developed ITV model have provided evidence that pathways *via* C_5 hydrocarbons, *i.e.* cyclopentadiene (C_5H_6) and the cyclopentadienyl radical (C_5H_5), are of great importance for the aromatics formation in the 1-pentene flame, because both species are easily formed from initial fuel decomposition products such as C_5H_8 and C_3H_5.

As described above, detailed reaction pathways from 1,3-pentadiene to 1,3-cyclopentadiene were adopted from the Aramco 3.0 mechanism [52] and incorporated into the newly developed ITV mechanism. Here, 1,3-pentadiene is mainly consumed by H-atom abstraction leading to the 1,3-pentadien-5-yl radical. This radical then mesomerizes to the 1,4-pentadien-3-yl radical, which then in turn mainly undergoes a cyclization reaction forming cyclopentadiene (C_5H_6). By including this detailed reaction path, which is highlighted in Fig. 7, the profiles of 1,3-cyclopentadiene and its direct product, the cyclopentadienyl radical, are well predicted regarding the peak mole fraction and the profile shape (see Fig. 8a-b). It should be emphasized that the accurate prediction of 1,3-cyclopentadiene and the cyclopentadienyl radical mole fractions is very important for the prediction of PAH formation, which will be highlighted in the following. While the aforementioned reaction channel $C_5H_8 \rightarrow C_5H_7 \rightarrow C_5H_6$ (cyclic) is not included in the NUIG_Nara mechanism, the USC mechanism also includes a pathway to describe the formation of 1,3-cyclopentadiene from 1,3-pentadiene, as evident from Fig. 7. First, the decomposition of 1,3-pentadiene yields C_5H_7 radicals by H-atom abstraction, and the latter can form 1,3-cyclopentadiene by releasing the second H-radical. While the first step contributes with 27% to a large part to the consumption of 1,3-pentadiene, the second step is only of minor importance, as C_5H_7 preferentially decomposes *via* β-scission in the USC mechanism. This leads to a strong under-prediction of the mole fraction of 1,3-cyclopentadiene, as shown in Fig. 8a.

When using the NUIG_Nara mechanism, C_5H_6 and C_5H_5 are strongly under-predicted, as shown in Fig. 8. From the above discussion, it is evident that the reaction sequences involving allylic bond breakage and describing the conversion from 1,3-pentadiene to 1,3-cyclopentadiene deserve particular attention, and the new ITV mechanism takes these pathways into account.

Mole fraction profiles of C_3H_3, C_6H_6 and C_7H_8 are depicted in Fig. 9. The propargyl radical (C_3H_3) as an important benzene precursor is detected with a peak concentration of 5.6×10^{-4} (Fig. 9a), which is slightly over-predicted by the newly developed ITV model. According to this model, propargyl radicals are mainly formed *via* H-atom abstractions from propyne (30%) and allene (26%). These reactions mainly contribute to propargyl formation in the USC model, and a severe over-prediction of C_3H_3 is noted for this model. Reasonable agreement between experimental and modeling results is obtained for the NUIG_Nara mechanism, in which the aforementioned formation pathways are less important.

Propargyl is generally considered to dominate benzene formation *via* $C_3H_3 + C_3H_3 \rightleftharpoons C_6H_6$ (and phenyl+H) [13,14]. Furthermore, the reactions of $C_3H_5 + C_3H_3$ and i-$C_4H_5 + C_2H_2$ can form benzene

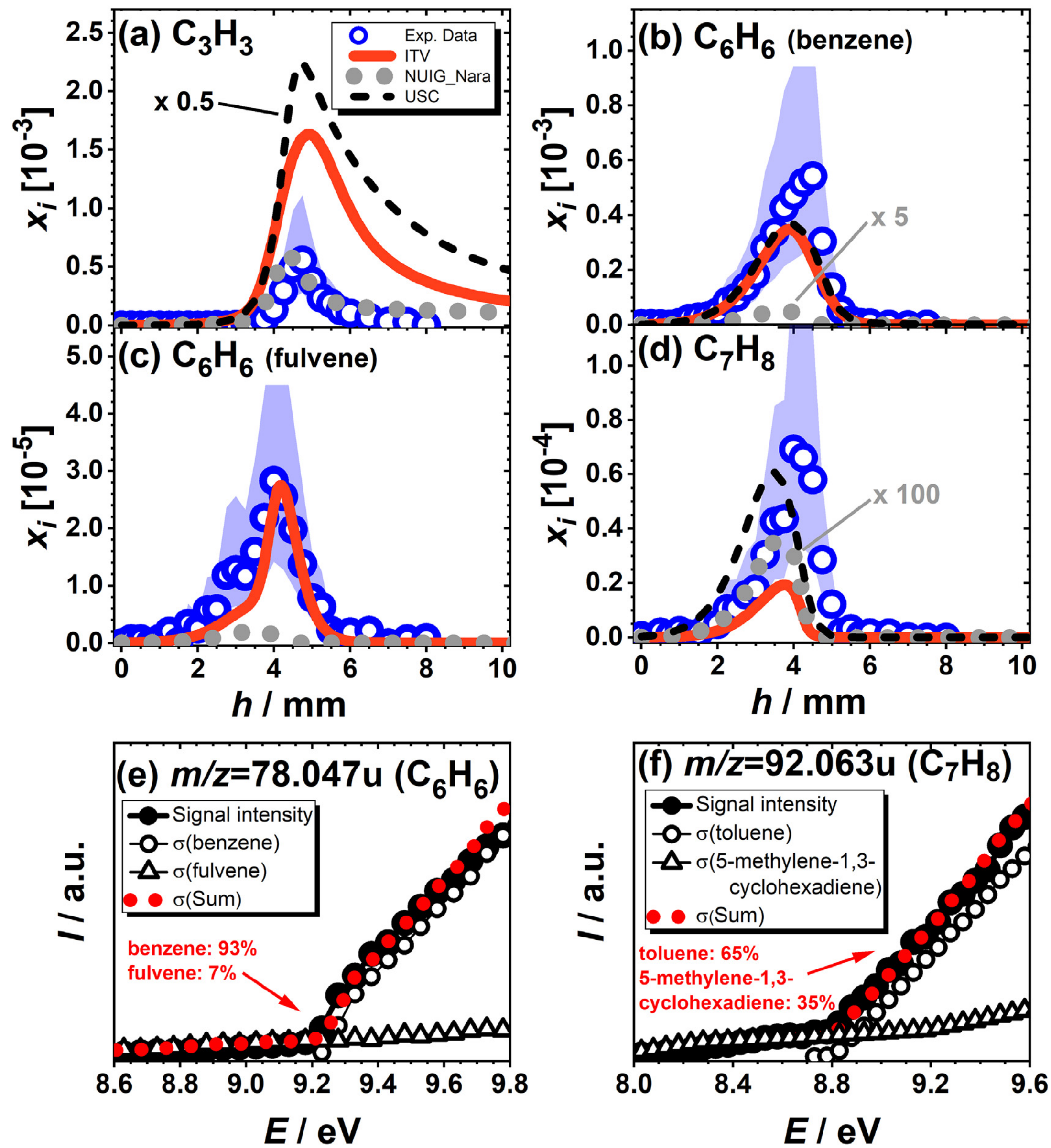

Fig. 9. Mole fractions x_i as function of the height above the burner h of (a) C_3H_3, (b) C_6H_6 (benzene), (c) C_6H_6 (fulvene), and (d) C_7H_8 (toluene). Symbols: experimental results; solid, dotted and dashed lines: simulations with the ITV, NUIG_Nara and USC models, respectively. Experimental uncertainties (factor of 2) are visualized as shaded area. Flame-sampled photoionization efficiency curves as function of the photon energy E for (e) $m/z = 78.047$ u (C_6H_6), and (f) $m/z = 92.063$ u (C_7H_8). Closed symbols (connected to guide the eyes) represent the measured signal, open symbols the tabulated photoionization cross section curve of the respective isomers, and the obtained sum signal is shown as dotted line. The contributions of the main isomers are shown as percentages.

via fulvene [55,58]. A high C_6H_6 peak mole fraction of 5.4×10^{-4} is detected with benzene as the main isomer (Fig. 9b and e). This mole fraction is slightly higher than peak mole fractions in flames of other fuels at similar conditions (for details see [16] and references therein), which might be explained by the easy accessibility of the respective benzene-forming reactants in the 1-pentene decomposition and the favored formation of 1,3-cyclopentadiene (see also Fig. 7). A close match between experiment and modeling results from the new ITV model is observed for the benzene mole fraction. While its mole fraction predicted by the USC mechanism is within the experimental uncertainty, the formation of benzene is significantly under-predicted by the NUIG_Nara model.

For the ITV mechanism as well as the NUIG_Nara model, 98% and 79% of benzene are formed *via* fulvene. This is in accordance with the results of Hansen et al. [55] for a premixed 1-hexene flame that have highlighted the importance of benzene formation pathways *via* fulvene, which itself is almost exclusively produced by the reaction of propargyl (C_3H_3) and allyl radicals (C_3H_5). In the ITV model, the reaction of cyclopentadienyl and methyl radicals contributes with 99% dominantly to the fulvene formation, whereas fulvene is mostly yielded from propargyl recombination (68%) and the reaction of C_4H_5 radicals with acetylene (22%) in the NUIG_Nara model. As described above, the reaction channel from 1,3-pentadiene to 1,3-cyclopentadiene is missing in the NUIG_Nara model, which leads to a significant under-prediction of cyclopentadiene and consequently fulvene and benzene formation. In the USC mechanism, fulvene is not included and the propargyl recombination directly leads to benzene formation with 83%. However, it should be mentioned that, while the propargyl recombination reaction in the USC mechanism has a similar rate constant as that in the ITV mechanism, its backward reaction is excluded, which artificially increases the net production of benzene. This compen-

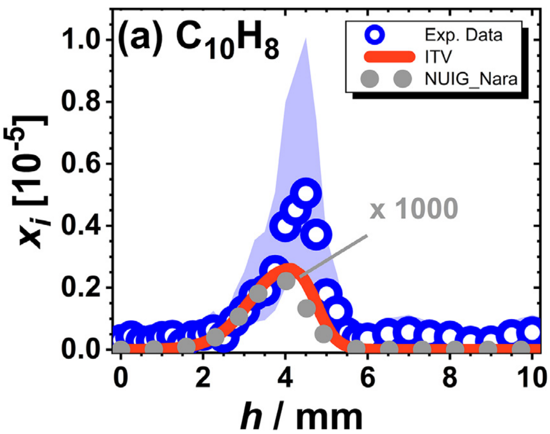

Fig. 10. Mole fraction x_i as function of the height above the burner h of $C_{10}H_8$. Symbols: experimental results; solid, dotted and dashed lines: simulations with the ITV and NUIG_Nara models, respectively. Experimental uncertainties (factor of 2) are visualized as shaded area.

sates the fact that the reaction channel from 1,3-pentadiene to 1,3-cyclopentadiene is missing in the USC mechanism and benzene and fulvene formation from C_5H_5 and C_5H_6 is thus inhibited.

According to the predictions of the ITV and NUIG_Nara model, benzene formation proceeds mainly *via* fulvene as an intermediate and this species is therefore of great importance for model validation. The experimental and simulated mole fraction profiles are shown in Fig. 9c. A close match between experiment and simulation regarding the profile shape and the peak mole fraction is observed for the newly developed ITV model, while the fulvene formation is under-predicted by the NUIG_Nara model. These strong deviations in fulvene formation can be traced back to differences in the dominant fulvene formation reactions, *i.e. via* C_5H_5 in the ITV and *via* C_3H_3 in the NUIG_Nara model (see Fig. 7). This observation again highlights the importance of the reaction channel $C_5H_8 \rightarrow C_5H_7 \rightarrow C_5H_6$ (cyclic) $\rightarrow C_5H_5$ (cyclic) that was mentioned in the beginning of this section and emphasized in Fig. 7.

In Refs. [16,59], the ratio of the peak mole fractions of toluene to benzene was compared for premixed flames of different fuels under similar flame conditions. It was shown that this ratio ranges between 0.11 and 0.21 for most of the fuels when the reaction $C_6H_5+CH_3 \rightleftharpoons C_7H_8$ dominates toluene formation. For the present 1-pentene flame, a ratio of 0.13 is experimentally observed, indicating that toluene formation mainly occurs *via* the aforementioned pathway. The reaction of $C_6H_5+CH_3 \rightleftharpoons C_7H_8$ is also the main contributor to toluene formation in the simulations with all models and similar reaction rate constants are used in the mechanisms. The major consumption channel of toluene is C_7H_8 $+H \rightleftharpoons C_7H_7+H_2$. The smaller rate constant of this reaction in the USC mechanism leads to a slower toluene consumption in the simulation. The experimental peak mole fraction of C_7H_8 (mainly toluene, Figs. 9d and f) of 6.9×10^{-5} is under-predicted by the ITV model. The NUIG_Nara model, which was, however, not specifically developed and validated for the high-temperature oxidation of 1-pentene, severely under-predicts the toluene formation by at least a factor of 100.

Further experimentally detected small aromatic species include naphthalene ($C_{10}H_8$), the experimental and simulated mole fraction profiles of which is presented in Fig. 10. Note that this species is not considered in the USC mechanism.

The experimental peak mole fraction of 5.0×10^{-6} for naphthalene and a close match between experiment and simulation is observed for the newly developed ITV model. For the NUIG_Nara model, however, the naphthalene mole fraction is under-predicted by a factor of near 1000. The main formation pathway in the ITV model is with 82% the recombination of two cyclopentadienyl radicals (see Fig. 7). While the reaction of cyclopentadienyl radicals to $C_{10}H_8$ is included in the NUIG_Nara mechanism, the formation of naphthalene is not important because of the low concentration of cyclopentadienyl radicals. Instead, $C_{10}H_8$ is formed *via* the reactions $C_9H_7+CH_3$ (41%) and $C_6H_5+C_4H_4$ (40%).

5. Conclusions

We have presented isomer-resolved mole fraction profiles measured in a fuel-rich laminar premixed low-pressure flame of 1-pentene using synchrotron-based PI-MBMS. The quantitative dataset was analyzed with numerical simulations performed with a newly developed model (here named as "ITV") and two models that had been taken ("USC") or combined ("NUIG_Nara") from the literature. The chemical structure of the premixed 1-pentene flame was inspected with a specific interest on the initial fuel decomposition and on formation pathways of small PAHs and their precursors.

Although both the NUIG [37] and the USC [36] mechanism were neither developed nor validated for 1-pentene, they contain this molecule as intermediate. These mechanisms were hence selected as a basis for the C_5 chemistry. The USC mechanism additionally includes reaction pathways of several aromatic species up to C_7H_8. In the NUIG_Nara version, the NUIG model was extended to allow for the description of growth reactions up to two-ring aromatics by incorporating the PAH chemistry taken from the model of Narayanaswamy et al. [38]. Although the fuel-specific reactions are included in the USC and NUIG base mechanisms, the predicted primary reaction pathways were found to vary remarkably, thus causing discrepancies in the predicted mole fraction profiles of small hydrocarbon intermediates. Moreover, the reaction pathways leading to aromatic species and small PAHs differ significantly between the mechanisms, and the models were not fully capable to describe the formation of growth species. The observed differences between the two literature models and the experimental data have motivated the development of the ITV mechanism – a new high-temperature 1-pentene mechanism, for which the experimental results here have provided guidance. The newly developed mechanism is based on the mechanism of Narayanaswamy et al. [38], which also includes a well-validated C_0-C_4 base mechanism [50]. The fuel-specific reactions of 1-pentene and their reaction rate parameters were taken from Refs. [18,36,37]. Overall, the simulations with the newly developed model were found to satisfactorily represent experimental mole fractions regarding the peak mole fractions and profile shapes. In combination with the experimental data it was shown that the C_5 chemistry plays an important role in formation reactions of growth species. In particular, it was highlighted that the reaction channel describing the conversion from 1,3-pentadiene to the cyclopentadienyl radical [$C_5H_8 \rightarrow C_5H_7 \rightarrow C_5H_6$ (cyclic)$\rightarrow C_5H_5$ (cyclic)], which was considered in the new ITV mechanism, is of great importance for the formation of aromatic hydrocarbons. Even though the new ITV model predicts the measured profiles of most species within the experimental uncertainty limits, further improvements of the mechanism might be considered in the future, for which the present data and other recent studies may provide valuable information. The ITV model can, for example, be further improved in terms of reaction pathways of large PAH molecules and pressure-dependent rate constants of sensitive reactions. For these modifications, however, additional experimental or theoretical efforts are required, which are beyond the scope of the present study. Such further investigations may be particularly useful in light of the important role of C_5 hydrocarbons as relevant components of realistic fuels such as liquefied natural gas (LNG) and as important intermediate species that are formed during technical relevant combustion processes.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2019.05.013.

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